YGGDRASIL
ENVIRONMENTAL SYSTEMS

OFFICIAL OPEN-CORE RELEASE · V1.0



CIRCULAR-CORE: OUROBOROS

Cyberphysical Material Metabolism Digital Twin, EWC/LER Gravimetric Accounting, Near-Infrared Polymer Metrology & Physics-Informed Multi-Quantile Scope 1–2–3 Decarbonization Platform

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CREDENTIALS: Environmental Engineer | Master's Degree in Environmental Engineering and Management (MECES Nivel 3 / EQF Level 7)

FRAMEWORK: EU CSRD (ESRS E1 & E5) & Ley 7/2022 / ISO 14064-1:2019, ISO 59020 & RD 1055/2022

REPOSITORY: github.com/Yggdrasil-Environmental-Systems/CIRCULAR-CORE-OUROBOROS

CERN/ZENODO: Reserved DOI: **10.5281/zenodo.24114014**

GOVERNANCE: EU AI Act (2024/1689) Art. 14/15 · ISO/IEC 42001 & ISO/IEC 27037

EXECUTIVE ABSTRACT (CIRCULAR GOVERNANCE, WASTE METABOLISM & CSRD DECARBONIZATION)

Healthcare networks, civil infrastructure projects, and industrial packaging operators across the European Union face unprecedented regulatory convergence under the Corporate Sustainability Reporting Directive (Directive EU 2022/2464, EFRAG ESRS E1 & E5), Spain's *Ley 7/2022* on Waste and Contaminated Soils, *Real Decreto 1055/2022* on Extended Producer Responsibility (EPR), and *Real Decreto 764/2026* governing telematic eSIR traceability. Conventional waste accounting relies on post-hoc invoicing estimates that obscure severe source-segregation distortions—most notably the systematic misrouting of ordinary municipal-like restroom paper (`LER 20 03 01`) into hazardous high-temperature biohazard incineration streams (`LER 18 01 03*`). **CIRCULAR-CORE: OUROBOROS** resolves these structural blind spots through a unified cyberphysical metabolism platform integrating high-frequency gravimetric/volumetric auditing, MITECO HP 1–HP 15 mirror-code hazard classification, Near-Infrared (NIR) optical polymer sorting, End-of-Waste (EoW) certification, LIFE-BIOBEST biowaste metrology, and ISO 14064-1:2019 corporate greenhouse gas accounting.

TECHNICAL & COMPUTATIONAL ABSTRACT (DUAL PYTORCH PINN-TCN & MASS-CONSERVATIVE METROLOGY)

OUROBOROS couples deterministic Material Flow Analysis (MFA, ISO 59020:2024) and ISO 14040/44 Life Cycle Assessment with a dual open-core PyTorch neural architecture: (1) `OuroborosAnomalyNet` (1,964 parameters, latent bottleneck $Z \in \mathbb{R}^4$, 0.40 ms CPU latency), an unsupervised LayerNorm-GELU autoencoder isolating unauthorized biohazard purges via weighted reconstruction MSE; and (2) `OuroborosCircularForecaster` (199,020 parameters, 3.87 ms CPU latency), a causal dilated WaveNet Temporal Convolutional Network with Bidirectional GRU exogenous encoding and 4-head Cross-Attention trained under `PhysicsInformedCircularLoss` to project strictly non-crossing P_{10}–P_{50}–P_{90} decarbonization trajectories. Evaluated on 25-month gravimetric benchmarks (3,501.91 kg across 23 units), OUROBOROS isolates a 92.74% restroom over-segregation paradox and demonstrates a -70.4% Scope 3 abatement (-83.2% total Market-Based corporate footprint) via steam sterilization (`D09`) and mechanical recycling (`R03`).

< 4 ms	-70.4%	134 Norms	92.74%
DUAL-AI LATENCY	**SCOPE 3 ABATEMENT**	**STATUTORY VAULT**	**ROOT-CAUSE ISOLATED**
0.40 ms AE · 3.87 ms PINN-TCN	Autoclave D09 + 30% R03 Credit	81 BOE · 28 DOUE · 25 ISO	Restroom LER Misclassification

Double DNA Armor Mandate: ATHENA and ASTRAEA generated the two master DNA capsules forming OUROBOROS's "double armor", expressly engineered to accompany the Core anywhere in the world (Git, Docker, Streamlit Cloud, or Hugging Face) up to the Open-Core boundary, certifying 134 immutable EU/Spanish/ISO standards, zero foreign jurisdictional leakage, and strict conservation of mass, stoichiometry, and thermodynamic laws.
(Registro Institucional Oficial: "ATHENA y ASTRAEA generaron las dos cápsulas maestras de ADN que forman la 'doble armadura' de OUROBOROS y que están diseñadas expresamente para seguir al Core a cualquier lugar del mundo (Git, Docker, Streamlit Cloud o AWS) hasta donde está el corte Open-Core").

1. INTERNATIONAL REGULATORY COMPLIANCE & DIRECTIVES

CIRCULAR-CORE: OUROBOROS enforces strict statutory compliance across European Union sustainability directives, Spanish circular economy legislation (BOE / MITECO), and international metrological standards (ISO / UNE-EN):

1.1. Circular Economy, EWC/LER & End-of-Waste

• **Ley 7/2022 & Decision 2014/955/EU (EWC/LER):** Governs the waste hierarchy, 6-digit European Waste Catalogue coding, and hazardous mirror-entry classification under Reg. (EU) 1357/2014 and Reg. (EU) 2017/997 (HP 1–HP 15).
• **RD 1055/2022, Reg. (EU) 2024/1157 & BOE-A-2026-21300:** Enforces closed packaging plant mass balances (≤ ±1.50% tolerance), UNE-EN 15343–15348 secondary polymer traceability, and thermoplastic End-of-Waste (EoW) criteria.
• **RD 105/2008 (RCD) & Decreto 83/1999 CAM:** Mandates ≥ 70% construction waste valorization and Class I–IV biosanitary segregation protocols.

1.2. Carbon Accounting, Biowaste & AI Governance

• **Directive (EU) 2022/2464 (CSRD) & UNE-EN ISO 14064-1:2019:** Standardizes ESRS E1 (Climate) and ESRS E5 (Resource Inflows/Outflows) inventories across Scopes 1, 2 (Location/Market-based), and 3 with 95% Gaussian uncertainty.
• **ISO 59020:2024, ISO 14040/44 & LIFE-BIOBEST (RD 506/2013):** Governs circularity boundary metrology, LCA substitution credits, and Class A/B/C compost purity (< 0.50% physical impurities d.m.; $AT_4 \leq 10$ mg O_2/g VS).
• **EU AI Act (Reg. 2024/1689) & ISO/IEC 42001 / 27037:** Enforces Art. 14/15 physical guardrails, anti-greenwashing clamping, and SHA-256 digital evidence sealing.

ACCOUNTING PARADIGM	COMPUTATIONAL LATENCY	MASS & POLYMER METROLOGY	SCOPE 3 & UNCERTAINTY ENGINE	STATUTORY & AI COMPLIANCE
Manual Ledger / Contractor Invoices	Days / Weeks (Post-Hoc)	Unverified Tare; Blind to Restroom Misrouting	Spend-Based Proxies; Zero Error Propagation	High Audit & Greenwashing Exposure
Unconstrained Black-Box AI	< 10 ms (Fast)	Violates Mass Continuity ($\Delta m \neq 0$)	Quantile Crossing & Impossible Negative Mass	Non-Compliant (EU AI Act Art. 15)
CIRCULAR-CORE: OUROBOROS (Open-Core)	**< 4 ms (Real-Time)**	**EN 45501 (≤0.5%) + NIR SAM + HP1–HP15**	**Activity-Based LCA + ISO 14064-1 (k=1.96)**	**134 Verified Norms · Sentinel Sealed**

2. MATHEMATICAL FOUNDATION & GOVERNING EQUATIONS

The OUROBOROS Open-Core engine couples deterministic gravimetric conservation, optical spectroscopy, and Gaussian metrology with physics-informed deep temporal learning:

2.1. Dynamic MFA Continuity & Cylindrical Volumetry

Conservation of mass across sorting/treatment nodes (ISO 59020 / EN 45501 Class III) and in-situ cylindrical container volumetry follow:

$$\Sigma\, m_{in}(t) = \Sigma\, m_{R03\text{–}R05} + \Sigma\, m_{D01\text{–}D10} + m_{evap} + \Delta S, \qquad V = \pi\,(d/2)^2\,\Delta h$$

Requiring scale closure $|\Delta m|/m_{in} \leq 0.5\%$ (plant mass balance ≤ $\pm 1.50\%$ under RD 1055/2022) and Dulong-Boie moisture correction $LHV_{ar} = LHV_{dry}(1 - W) - 2.442\,W$.

2.2. Scope 1–2–3 Inventory & ISO 14064-1 Uncertainty

Value-chain GHG emissions integrate treatment stoichiometry, road freight ($t{\cdot}km$), virgin HDPE bags (28 g), and 95% Gaussian error propagation:

$$E_{S3} = \Sigma\, m_k\, EF_k + \Sigma\,(m_k\, L_k)\, EF_{tkm}, \qquad U_{95} = 1.96\,\sqrt{\Sigma\,(E_i \cdot u_i)^2}$$

With MITECO/IPCC AR6 factors: incineration *1.80*, steam autoclave *0.22*, landfill *0.58*, and mechanical recycling credit *-0.70 kg CO_2e/kg*.

2.3. NIR Spectral Angle Mapper & HP 14 Ecotoxicity

Optical discrimination across 22 NIR wavelengths (900–1750 nm) and Reg. (EU) 2017/997 aquatic ecotoxicity mirror classification satisfy:

$$\theta_{SAM} = \arccos\left[\,(u \cdot v)\,/\,(||u||\,||v||)\,\right], \qquad I_{HP14} = 100\Sigma c_{H410} + 10\Sigma c_{H411} + \Sigma c_{H412} \geq 25\%$$

Where $\theta_{SAM} \to 0$ identifies monomaterial resins vs. EVOH/PA barrier multilayers (1420/1680 nm absorption peaks) for pneumatic ejection.

2.4. Multi-Quantile PINN-TCN Composite Loss

`OuroborosCircularForecaster` minimizes asymmetric pinball loss regularized by non-crossing, non-negativity, and stoichiometric bounds:

$$L = L_{pinball} + 1.5\,\mathrm{ReLU}(P_{10}\text{-}P_{50}) + 1.5\,\mathrm{ReLU}(P_{50}\text{-}P_{90}) + 2.0\,\mathrm{ReLU}(-\hat{y}) + 0.5\,L_{ratio}$$

Where L_{ratio} constrains implied emission intensity $\hat{y}_{CO2}/\hat{y}_{mass} \in [0.05, 3.00]$ *kg CO_2e/kg* across all 12 forecast quarters.

3. SYSTEM ARCHITECTURE, SENTINEL & DOUBLE DNA ARMOR

CIRCULAR-CORE: OUROBOROS implements a hexagonal cyberphysical architecture separating deterministic statutory engines from probabilistic PyTorch neural operators, ensuring full offline reproducibility across local workstations, Docker containers, and cloud spaces.

3.1. Hexagonal Cyberphysical Architecture & Dual PyTorch Neural Lab

The Open-Core platform orchestrates three decoupled layers: (1) **Deterministic Metrology & LCA Layer** (`core/`), executing gravimetric/volumetric balances, MITECO 2021 mirror-code decision trees, NIR SAM polymer classification, EoW certification, LIFE-BIOBEST compost auditing, and Scopes 1–2–3 ISO 14064-1 inventories; (2) **Dual PyTorch Neural Layer** (`neural_lab/models.py`, 200,984 combined parameters), pairing `OuroborosAnomalyNet` (8→32→16→4 autoencoder) with `OuroborosCircularForecaster` (4-layer causal WaveNet TCN with dilations $d = 1, 2, 4, 8$, BiGRU exogenous encoder, and 4-head Cross-Attention); and (3) **Cryptographic Governance Layer** (`compliance/auto_verificador_integridad.py`), verifying SHA-256 capsule integrity prior to execution.

> **Anti-Hallucination & Anti-Greenwashing Circuit-Breaker:** In compliance with EU AI Act (Reg. 2024/1689 Art. 15) and ISO/IEC 42001, OUROBOROS executes two concurrent runtime interceptors: `AIGuardrailEngine` (`core/ai_guardrails.py`) and `PhysicsSentinelSidecar` (`physics/sentinel_sidecar.py`). Any neural prediction exhibiting negative mass ($m < 0$), statistical excursion above $\mu + 4\sigma$, incineration intensity below stoichiometric minimums (< 0.80 kg CO_2e/kg), or exaggerated circular recycling credits beyond -1.50 kg CO_2e/kg is instantaneously clamped and logged with an immutable SHA-256 audit seal.

3.2. Athena & Astraea Double DNA Armor

To guarantee self-contained statutory verification and physical consistency up to the Open-Core boundary, OUROBOROS embeds two cryptographically sealed JSON sidecars:

1. Astraea Regulatory DNA (`compliance/regulatory_dna.json`) Generated by **VERI-AUDIT: ASTRAEA**, this statutory vault encodes **134 verified norms** (81 Spain BOE, 28 European Union DOUE, 25 International ISO/UNE-EN) with 0% non-EU/Spanish leakage (CSRD, Ley 7/2022, RD 1055/2022, RD 764/2026, Reg. 2017/997, ISO 14064-1, ISO 59020, EU AI Act), sealed under ISO/IEC 27037 (SHA-256: `1b26e8ce8097...`).

2. Athena Scientific DNA (`physics/scientific_dna.json`) Engineered by **AI-CORE: ATHENA**, this mathematical vault codifies **4 immutable physical invariants** (`INV_OURO_01` to `04`: EN 45501 mass closure, Dulong-Boie LHV, ISO 14064-1 quadratic uncertainty, and EoW purity), **6 clamping bounds**, and **12 peer-reviewed references** (Cambridge MFA, CNRS Dynamic LCA, LIFE-BIOBEST), sealed via SHA-256 (`d8daf13bf591...`).

3.3. Scalable Technology Roadmap (Open-Core to Enterprise Suite)

GENERATION	OPERATIONAL SCOPE	COMPUTATIONAL & STATUTORY SPECIFICATION	DEPLOYMENT & LICENSING
OUROBOROS V1 (Open-Core)	Scientific, LCA & Circular Baseline (Current)	Scopes 1–2–3 & ISO 14064-1 Engine + Gravimetric/Volumetric MFA + MITECO HP1–HP15 + NIR SAM & EoW + LIFE-BIOBEST + Open-Core Dual PyTorch (`OuroborosAnomalyNet` 1,964p & `OuroborosCircularForecaster` 199,020p) + Double DNA Armor.	Open-Core / CERN Zenodo Archive (GPLv3 / CC BY-NC-SA 4.0)
OUROBOROS V2 (Enterprise Suite)	Corporate ESG, Big 4 Assurance & eSIR	• **TITAN Industrial Neural Weights (NVIDIA L4):** `OuroborosDeepLatentAE` ($Z \in \mathbb{R}^{16}$, Residual Highway blocks, 276,120 params) & `OuroborosDeepPINNForecaster` (6 WaveNet layers, 8-head Attention, 7-Quantile P_{05}–P_{95} architectural monotonicity, 1,518,718 params). • **Ley 7/2022 Fiscal & SCRAP Tax Engine:** Automated liquidation of landfill tax (40 €/t), incineration tax (15 €/t), non-reusable plastic levy (0.45 €/kg), and RD 1055/2022 eco-modulation. • **MITECO RD 764/2026 & E3L v3.0 XML Exporter:** 3-tier 10-digit NIMA validation, Orden TED/1024/2026 biunivocal LER→R/D admissibility blocking, and telematic eSIR DI generation. • **RD 105/2008 Municipal Bond & CSRD Suite:** Civil works ≥70% valorization bond release certificates, multi-site corporate benchmarking, and ESRS E1/E5 Big 4 limited-assurance tables.	Commercial Enterprise License / Healthcare Consortia, EPC Contractors & ESG Auditors
OUROBOROS V3 (Planetary)	Industrial Symbiosis & DPP Mesh	Cross-sector secondary raw material exchange coupled with EU Digital Product Passports (ESPR Reg. 2024/1781) and inline conveyor hyperspectral edge inference.	European Circular Industrial Infrastructure Sovereign Network

4. GRAVIMETRIC METABOLISM, ANOMALY AUDIT & OVER-SEGREGATION

Longitudinal gravimetric auditing on **Protocol ALPHA** (25 continuous months, 2024-03 to 2026-03, 3,501.91 kg total across 23 functional units) demonstrates how coupling physical mass accounting with unsupervised deep autoencoding exposes both episodic operational purges and chronic structural misclassifications.

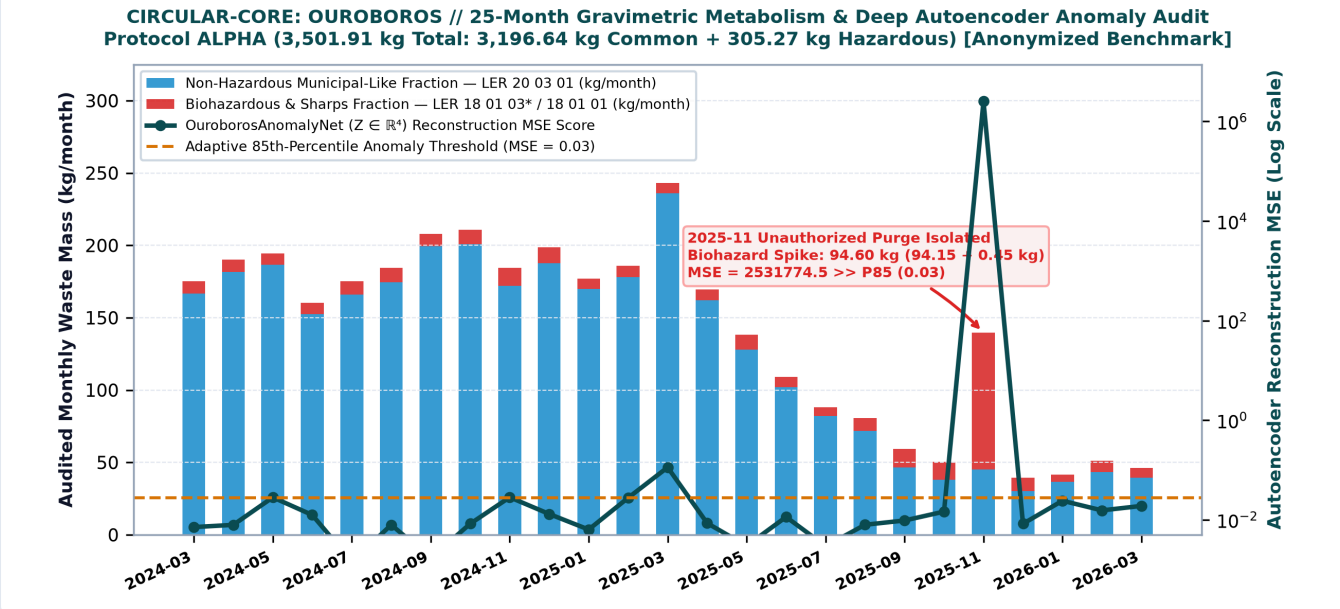


Figure 1: 25-Month Gravimetric Metabolism & Unsupervised Deep Autoencoder Audit (Protocol ALPHA). Monthly mass balance separating municipal-like fractions (3,196.64 kg total, 91.28%) from hazardous biohazard/sharps streams (305.27 kg total, 8.72%). OuroborosAnomalyNet ($Z \in \mathbb{R}^4$, 1,964 parameters) compresses 8 operational indicators and automatically isolates the Month-21 (2025-11) anomalous warehouse purge (94.15 kg biohazard + 0.45 kg sharps = 94.60 kg vs. ~8.5 kg/month baseline) via reconstruction MSE exceeding the 85th-percentile threshold by two orders of magnitude.

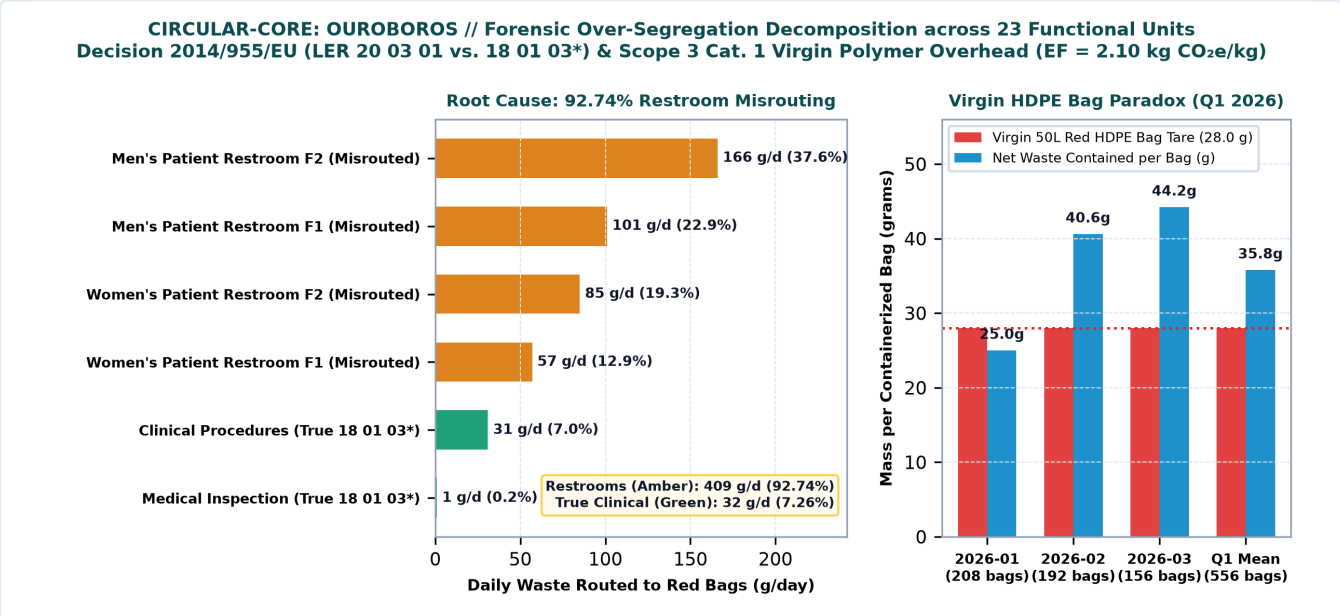


Figure 2: Forensic Decomposition of the 92.74% Restroom Over-Segregation & Virgin HDPE Bag Paradox. Left: Granular audit across all 23 units reveals that 0.409 kg/day (92.74%) of waste routed into hazardous red bags (LER 18 01 03*) originates from 4 ordinary outpatient restrooms (hand-drying paper towels properly classified as non-hazardous LER 20 03 01 under Decision 2014/955/EU), while true clinical biohazard accounts for only 0.032 kg/day (7.26%). Right: In Q1 2026, net waste per 50L red bag averaged 35.8 g (dropping to 25.0 g in Jan 2026) against a 28.0 g virgin HDPE bag tare (EF = 2.10 kg CO_2e/kg), proving nearly as much virgin polymer was incinerated as clinical waste.

5. ANONYMIZED MULTI-SECTOR BENCHMARK & STRESS-TESTING

ANONYMIZED BENCHMARK & LEGAL NON-AFFILIATION DISCLAIMER: All facility identifiers, NIMA registration codes (e.g., `2800000101`), corporate titles, civil works contracts, packaging sorting batches, and compost laboratory assays presented in this study constitute **GDPR/ISO 27001-anonymized and synthetic calibration benchmark protocols** engineered strictly to validate the mathematical, spectroscopic, and statutory engines of CIRCULAR-CORE: OUROBOROS. They do **NOT** represent official administrative inspection records, commercial audit disclosures, or legal non-conformities of any named public hospital, municipality, or private operator.

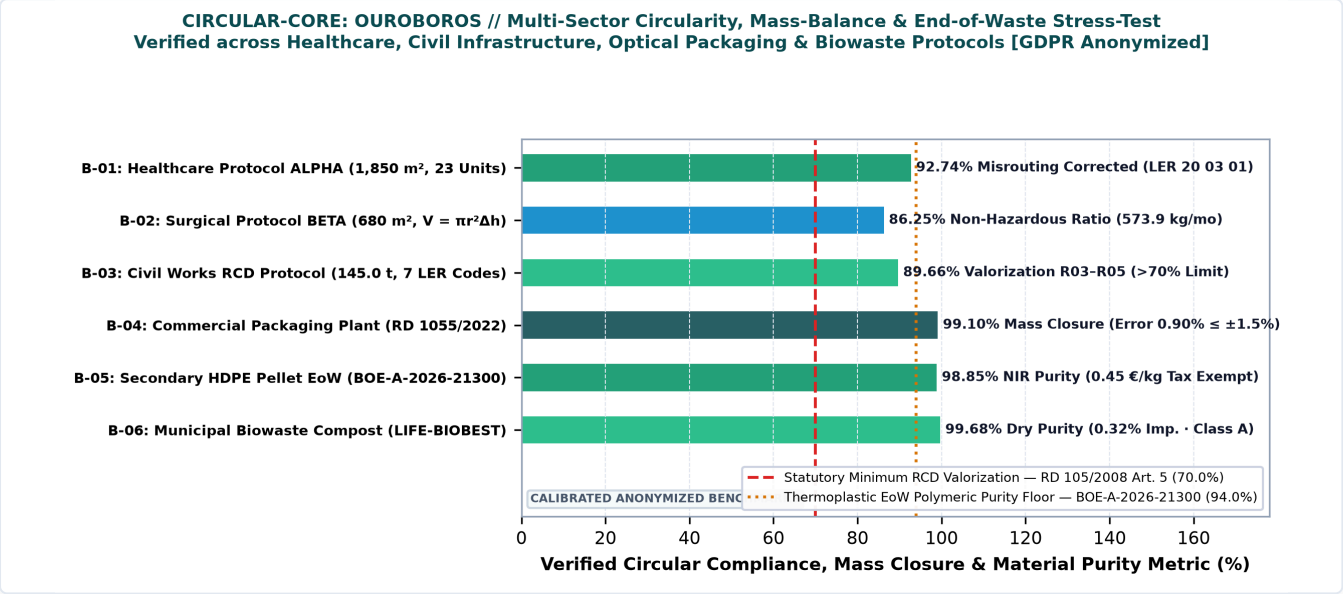


Figure 3: Multi-Sector Circularity, Mass-Balance Closure & End-of-Waste Stress-Test. Cross-sector validation spanning outpatient healthcare (Protocol ALPHA), invasive surgical/dental care (Protocol BETA), civil infrastructure demolition/excavation (Civil Works RCD), RD 1055/2022 optical packaging sorting, BOE-A-2026-21300 secondary HDPE pelletization, and LIFE-BIOBEST Class A municipal biowaste composting.

5.1. Multi-Sector Benchmark Matrix & Statutory Verification

Summary of verified gravimetric balances, hazardous mirror-code evaluations, and statutory thresholds across the 6 open-core calibration protocols:

BENCHMARK PROTOCOL	AUDITED FLOW / SCALE	DOMINANT LER & PATHWAY	KEY METROLOGICAL METRIC	STATUTORY VERDICT
B-01: Healthcare ALPHA	3,501.91 kg (25 Months · 23 Units)	20 03 01 / 18 01 03* → D09 + R03	**92.74%** Restroom Over-Segregation	Re-routed to Non-Hazardous; -70.4% S3
B-02: Surgical BETA	573.9 kg/mo (680 m² · V=πr²Δh)	18 01 03* (73.5) / 18 01 01 (5.4 kg)	**13.75%** True Invasive Biohazard	Compliant Decreto 83/1999 Class III
B-03: Civil Works RCD	145.0 t (7 LER Fractions)	17 05 04 / 17 01 01 → R05 (130.0 t)	**89.66%** Valorization (≥70% Min.)	RD 105/2008 Compliant · Bond Eligible
B-04: Packaging Plant	10,000 kg Commercial Lot	15 01 02 Polyolefins → R03 / R01	**0.90%** Mass Discrepancy (≤±1.50%)	RD 1055/2022 & Reg. 2024/1157 Valid
B-05: Secondary HDPE EoW	10,000 kg Extruded Pellet Batch	UNE-EN 15344 / ISO 1133-1 MFI	**98.85%** NIR Purity · 0.22% Moisture	EoW Certified · 0.45 €/kg Tax Exempt
B-06: Biowaste Compost	Municipal Organic Fraction	20 01 08 → R03 Aerobic Stabilization	**0.32%** Impurities · AT_4=6.2 · C/N=15.5	LIFE-BIOBEST & RD 506/2013 Class A

6. NIR SPECTRAL METROLOGY & ISO 14064-1 DECARBONIZATION

Achieving legally defensible End-of-Waste status and verified corporate decarbonization requires coupling inline optical polymer sorting (OUROBOROS SPECTRA) with activity-based Scope 1–2–3 accounting and quadratic uncertainty propagation.

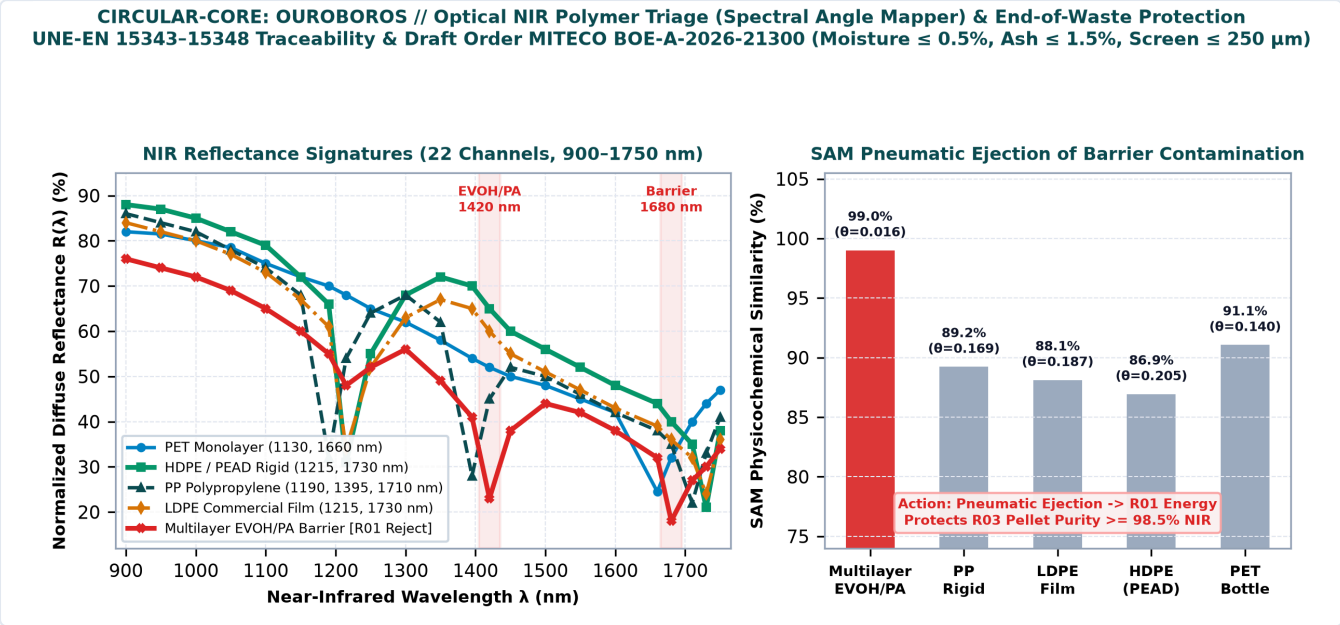


Figure 4: Near-Infrared (NIR) Diffuse Reflectance Signatures & Spectral Angle Mapper (SAM) Barrier Ejection. Left: Calibrated 22-wavelength reflectance curves (900–1750 nm) across 5 packaging polymer families, highlighting C–H overtone absorption valleys for PET (1130, 1660 nm), HDPE/LDPE (1215, 1730 nm), PP (1190, 1395, 1710 nm), and incompatible EVOH/PA multilayer barriers (1420, 1680 nm). Right: Under optical sensor noise, SAM isolates multilayer barrier contamination at 99.0% angular similarity ($\theta = 0.016$ *rad*), triggering pneumatic ejection to R01 energy recovery and preserving R03 pellet purity for BOE-A-2026-21300 End-of-Waste certification.

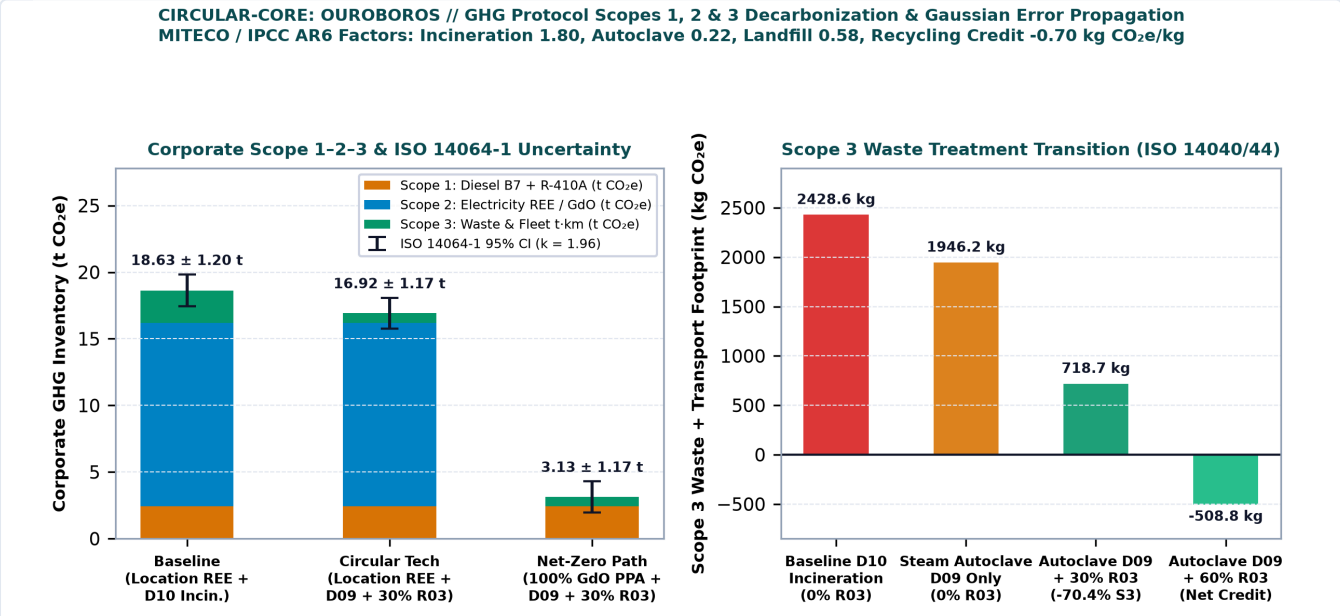


Figure 5: Corporate GHG Inventory (UNE-EN ISO 14064-1:2019) & Scope 3 Technological Abatement Sensitivity. Left: Protocol ALPHA consolidated inventory combining Scope 1 (2.42 t CO_2e: 420 L diesel B7 + 0.65 kg R-410A fugitive recharge at AR6 GWP = 2,088), Scope 2 (13.78 t CO_2e Location-Based REE vs. 0.00 t CO_2e Market-Based 100% GdO), and Scope 3, reducing total footprint from **18.63 ± 1.20 t CO_2e** to **3.13 ± 1.17 t CO_2e** (-83.2%). Right: Transitioning hazardous waste from 1,200 °C incineration (D10) to steam autoclave sterilization (D09) plus 30% mechanical recycling (R03) cuts Scope 3 emissions by **-70.4%** (from 2428.6 to 718.7 kg CO_2e), reaching net negative circular credit (-508.8 kg CO_2e) at 60% recycling.

7. MULTI-QUANTILE PINN FORECASTING & FORENSIC OPINION

To support multi-year CSRD transition plans (ESRS E1-4 decarbonization targets) without exposing sustainability officers to unphysical neural extrapolations, `OuroborosCircularForecaster` projects quarterly probabilistic envelopes governed by strict thermodynamic guardrails.

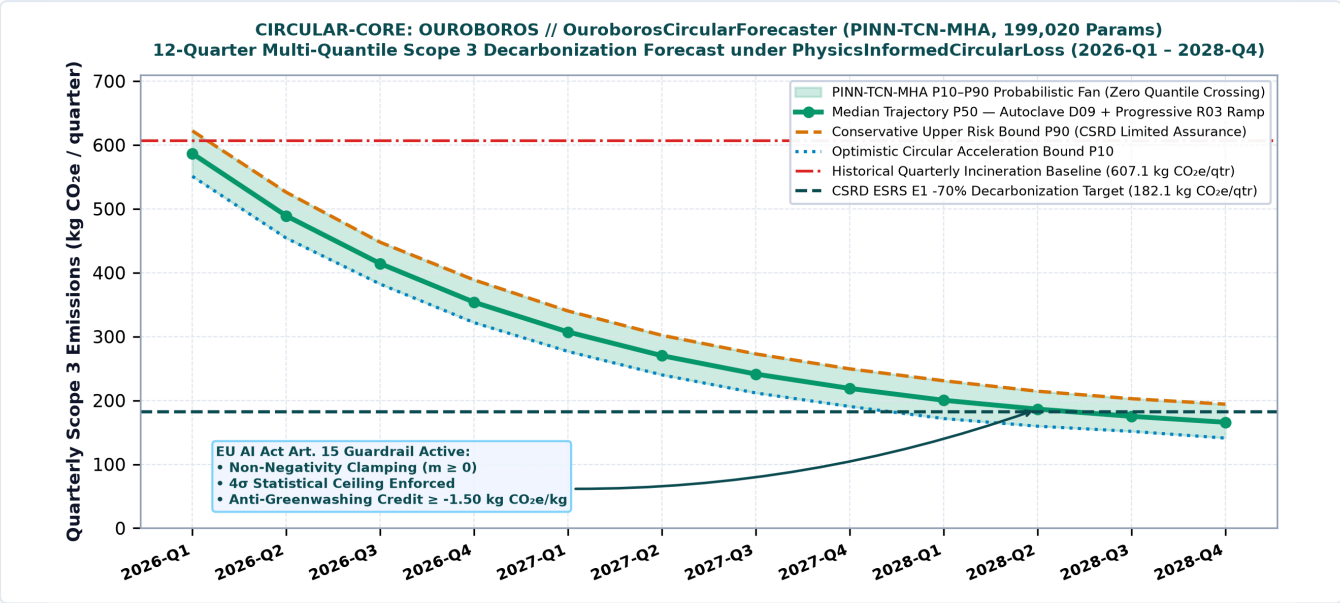


Figure 6: 12-Quarter Probabilistic Scope 3 Decarbonization Trajectory (2026-Q1 to 2028-Q4). Generated by `OuroborosCircularForecaster` (199,020 parameters; 4 dilated causal WaveNet blocks + BiGRU + 4-head Cross-Attention) trained under `PhysicsInformedCircularLoss`. Progressive enforcement of outpatient restroom segregation (`LER 20 03 01`), steam sterilization (`D09`), and mechanical recycling (`R03` ramp to 45%) drives median P_{50} emissions across the CSRD -70% threshold (182.1 kg CO_2e/quarter) by 2028-Q3 with zero quantile crossing ($P_{10} \leq P_{50} \leq P_{90}$).

FORENSIC ENGINEERING OPINION (EXPERT OPINION UNDER ART. 335.2 LEC) SPANISH CIVIL PROCEDURE ACT (LEY DE ENJUICIAMIENTO CIVIL)

Expert Declaration under Solemn Affirmation: The undersigned, in his professional standing as Environmental Engineer and Master in Environmental Engineering and Management (MECES Nivel 3 / EQF Level 7), issues this technical engineering opinion under strict adherence to the principles of scientific objectivity, independence, and metrological rigor prescribed in Article 335.2 of the Spanish Civil Procedure Act (*Ley 1/2000, de Enjuiciamiento Civil - LEC*).

Forensic Diagnosis of Circular & Fiscal Distortion: Empirical gravimetric decomposition demonstrates that applying obsolete facility-wide biohazard classifications in outpatient healthcare settings distorts hazardous mass accounting by up to 92.74%, causing harmless sanitary paper (`LER 20 03 01`) and under-filled virgin HDPE bags (28.0 g tare vs. 35.8 g net waste) to undergo high-emission thermal destruction (1.80 kg CO_2e/kg). Analogously, unverified polymer sorting in commercial packaging streams risks breaching the ±1.50% mass-balance ceiling under RD 1055/2022 and forfeiting the 0.45 €/kg plastic tax exemption under Ley 7/2022 Art. 72.

The coupling of EN 45501 gravimetric continuity, MITECO HP 1–HP 15 mirror-code verification, NIR Spectral Angle Mapper triage, ISO 14064-1:2019 Gaussian uncertainty propagation, and EU AI Act Art. 15 guarded PINN forecasting within **CIRCULAR-CORE: OUROBOROS** establishes a reproducible, cryptographically sealed (ISO/IEC 27037) evidentiary standard suitable for third-party ESG assurance and administrative compliance.

8. OPEN SCIENCE DECLARATION & MODULAR LICENSING

To advance transparent circular economy metrology, reproducible peer review, and verifiable decarbonization science while protecting the moral authorship and intellectual property of the author (Kelvin Jesús Flores Yarihuaman), CIRCULAR-CORE: OUROBOROS is published under a **Tri-Modular Open-Core Licensing Scheme**:

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PERMANENT ARCHIVAL DEPOSIT (CERN / ZENODO)
Reserved DOI: 10.5281/zenodo.24114014
Target Resolver: https://doi.org/10.5281/zenodo.24114014
Open-Core Repo: github.com/Yggdrasil-Environmental-Systems/CIRCULAR-CORE-OUROBOROS
Scan to verify open-access source archive, DNA capsules, and benchmarks.

KELVIN JESÚS FLORES YARIHUAMAN
Lead Systems Architect & Environmental Engineer
Master's Degree in Environmental Engineering & Management
MECES Nivel 3 / EQF Level 7

Yggdrasil Environmental Systems
Electronic Archival Record Verified · CERN / Zenodo

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